ECG Signal Denoising Using Deep Learning

A Comparative Study of Classical and Deep Learning Approaches

Capstone Project Report

Methods Benchmarked: Classical · Fourier · Wavelet · DRNN · FCN-DAE · 1D U-Net · CNN-BiGRU

1. Abstract

Electrocardiogram (ECG) signals are routinely degraded by real-world artifacts including baseline wander (BW), muscle activation noise (MA), and electrode motion (EM), which critically compromise downstream clinical analysis and automated diagnosis pipelines. This project addresses the challenge of ECG signal denoising by systematically benchmarking seven methods spanning classical signal processing and modern deep learning architectures.

The methods evaluated include: (1) Classical bandpass filtering, (2) Fourier Transform-based filtering, (3) Discrete Wavelet Transform (DWT) thresholding, (4) Deep Recurrent Neural Network (DRNN), (5) Fully Convolutional Denoising Autoencoder (FCN-DAE), (6) 1D U-Net, and (7) a hybrid CNN-BiGRU model. All deep learning models were trained on the MIT-BIH Arrhythmia Database and the QT Database, with noise sourced from the MIT-BIH Noise Stress Test Database (NSTDB) covering all three major artifact types.

Evaluation was conducted using Percentage Root Mean Square Difference (PRD), Root Mean Square Error (RMSE), Signal-to-Noise Ratio (SNR), and SNR Improvement (SNRimp). The proposed CNN-BiGRU model achieved the highest SNR improvement of +16.96 dB among all deep learning models, with a PRD of 27.78% and RMSE of 0.0753, closely rivalling the FCN-DAE baseline while employing a significantly lighter architecture. Wavelet denoising achieved the best PRD overall (1.21%), but lacks the adaptability and generalizability of data-driven deep learning approaches.

2. Related Work

The following 35 research papers form the literature basis for this project, organized thematically.

2.1 FCN-DAE & Denoising Autoencoders

References [1–5, 7–8]: Covers the original FCN-DAE (Chiang et al., 2019), RunDAE, FGDAE, attention-based ACDAE, and CBAM-DAE architectures.

2.2 DRNN & Recurrent Denoising

References [2–4, 34]: Antczak's original DRNN (2018), the biGRU denoiser (Dias et al., 2024 — base paper), and stacked CNN-based cardiac cycle denoisers.

2.3 CNN & Transformer Denoising

References [9, 10, 14, 15]: Multi-scale patch + Transformer denoising, APtrans-CNN, and the 1D CNN survey by Kiranyaz et al.

2.4 1D U-Net & Encoder-Decoder Architectures

References [11–13, 33]: Lightweight 1D U-Net (Miao et al., 2023), DeepFilter for baseline wander, UNet++ skip connections, and a comprehensive encoder-decoder survey.

2.5 Wavelet & Classical Methods

References [17–20, 35]: DWT thresholding, Variational Mode Decomposition (VMD), Empirical Mode Decomposition (EMD), and EEMD with genetic algorithm thresholding.

2.6 BiGRU / GRU Architectures for ECG

References [21–24]: BiGRU-based biometric identification (Lynn et al., 2019), CNN+attention-BiGRU, multi-stream BiGRU, and SE-BiGRU.

2.7 Hybrid CNN-RNN/GRU

References [25–27]: CNN+BiLSTM classification, CNN+BiGRU+attention (CBGM, Bai et al., 2024), and deep CNN+BiLSTM.

2.8 GAN-Based Denoising

References [28–29]: Adversarial ECG denoising (Wang et al., 2019) and multichannel GAN-based high-noise denoiser.

2.9 Surveys, Reviews & Datasets

References [16, 30–33]: PhysioNet/MIT-BIH (Goldberger et al., 2000), DL-ECG survey 2017–2023, JMIR systematic review, noise removal review, and encoder-decoder survey.

3. Base Paper

Selected Base Paper

> M. Dias, P. Probst, L. Silva, and H. Gamboa, "Cleaning ECG with Deep Learning: A Denoiser Tested in Industrial Settings," SN Comput. Sci., vol. 5, no. 5, p. 699, Jul. 2024, doi: 10.1007/s42979-024-03017-7.

Rationale for Selection

This paper was selected as the base reference because it is the closest direct architectural predecessor to our proposed model. Dias et al. employ a standalone bidirectional GRU (biGRU) for ECG denoising — which is precisely the recurrent backbone inside our CNN-BiGRU. The task (signal denoising), evaluation metrics (PRD, RMSE, SNR improvement), and problem domain are all identical, making this the most appropriate paper for a meaningful comparison.

Base Paper Summary

- Proposes a biGRU-only denoiser (no CNN front-end) trained to remove three real-world ECG artifacts: muscle activation (MA), electrode motion (EM), and baseline wander (BW).
- Training data: PTB-XL database (21,837 12-lead recordings) combined with noise from the MIT-BIH NSTDB — claimed to be the first work to combine these two databases.
- Achieves best RMSE of 0.023–0.041 and SNR improvements up to 30.6 dB depending on input SNR level.

- Additionally validated on real-world data from 43 industrial workers performing physical tasks — the first ECG denoising paper with such an industrial deployment evaluation.
- Compared only against two open-source denoisers from the Neurokit Python toolbox, with no comparison against FCN-DAE, DRNN, or 1D-UNet.

4. Datasets Used

4.1 MIT-BIH Arrhythmia Database

- Source: PhysioNet (Goldberger et al., 2000) — doi: 10.1161/01.CIR.101.23.e215
- Content: 48 half-hour two-lead ambulatory ECG recordings from 47 subjects, sampled at 360 Hz.
- Use in this project: Primary source of clean ECG windows for CNN-BiGRU training and testing. Yielded 59,002 training, 19,668 validation, and 19,668 test windows of length 512 samples.
- Standard benchmark in the ECG denoising literature, enabling direct comparison with published results.

4.2 QT Database

- Source: PhysioNet — 105 ECG recordings with detailed waveform annotations (P-wave, QRS, T-wave onset/offset).
- Sampled at varying frequencies; resampled to 360 Hz using polyphase resampling (resample_poly) in preprocessing.
- Use in this project: Source of clean ECG beats for DRNN and FCN-DAE training. Beats were extracted using P-wave start annotations, zero-padded to 512 samples, and DC-offset corrected.
- A fixed test set of 13 recordings (sel123, sel233, sel307, etc.) was held out for evaluation.

4.3 MIT-BIH Noise Stress Test Database (NSTDB)

- Source: PhysioNet — dedicated noise recordings of the three dominant ECG artifact types.
- Noise types: Baseline Wander (BW), Muscle Artifact (MA), Electrode Motion (EM).
- Use in this project: Real physiological noise was mixed into clean QT database beats for DRNN and FCN-DAE training using amplitude-adaptive scaling (alpha/beta/gamma coefficients). 13% of each noise channel reserved for test set.
- For CNN-BiGRU: Synthetic Gaussian noise (σ=0.5) was used instead, providing a controlled, reproducible noise floor.

5. Problems Identified in Base Paper

5.1 No Convolutional Front-End for Spatial Feature Extraction

Dias et al. feed the raw noisy signal directly into the biGRU without any convolutional preprocessing. This bypasses the opportunity to extract multi-scale local morphological features (QRS complex shape, P-wave and T-wave profiles) before temporal modelling begins. Raw signal input forces the GRU to simultaneously perform feature extraction and temporal dependency modelling, increasing the learning burden.

5.2 Dependency on 12-Lead Clinical ECG Setup

The base paper trains and validates exclusively on PTB-XL, which is a 12-lead clinical database. This restricts applicability to hospital-grade equipment and excludes the rapidly growing domain of single-lead wearable ECG devices (smartwatches, Holter monitors, fitness patches), which represent the most common real-world deployment scenario for automated ECG analysis.

5.3 Narrow Baseline Comparison (Only 2 Methods)

Dias et al. compare their biGRU denoiser only against two open-source tools available in the Neurokit Python library. No comparison is made against any deep learning baselines such as FCN-DAE, DRNN, 1D-UNet, or CNN-based autoencoders. This makes it impossible to situate the biGRU denoiser within the broader deep learning landscape.

5.4 No Reproducible Controlled Noise Evaluation

The base paper uses real physiological noise from the NSTDB at fixed SNR conditions, but does not report results across a systematically varied noise amplitude range. This makes it difficult to assess model robustness under different noise severities and complicates fair comparison with the broader literature which uses synthetic Gaussian noise at controlled amplitudes.

5.5 Computational Complexity Not Formally Quantified

While the paper claims its model has lower complexity than state-of-the-art approaches, no parameter count, FLOPs, or inference time measurements are reported. This makes it impossible to verify the efficiency claim or compare deployment feasibility on edge devices.

6. Our Proposed Solutions

6.1 Hybrid CNN-BiGRU Architecture

We address the missing convolutional front-end by prepending two 1D convolutional layers (kernel sizes 7 and 5) before the biGRU. The CNN block extracts multi-scale local temporal features from the raw noisy ECG, providing the biGRU with a structured feature representation rather than raw signal values. This

division of labour — CNN for spatial feature extraction, BiGRU for bidirectional temporal modelling — is the core architectural contribution relative to the base paper.

6.2 Single-Lead MIT-BIH Benchmark

We train and evaluate on the MIT-BIH Arrhythmia Database (single-lead), the most widely used benchmark in ECG denoising literature. This enables direct comparison with FCN-DAE, DRNN, and 1D-UNet under identical conditions, and demonstrates applicability to single-lead wearable devices — a deployment context entirely absent from the base paper.

6.3 Comprehensive Six-Method Benchmark

Rather than comparing against two baselines, we evaluate seven methods on the same dataset and noise conditions: Classical filtering, Fourier Transform, Wavelet (DWT), DRNN, FCN-DAE, 1D U-Net, and our CNN-BiGRU. This provides a complete picture of where each approach stands across PRD, RMSE, SNR, and SNR improvement metrics.

6.4 Controlled Gaussian Noise Baseline

We use synthetic Gaussian noise (σ=0.5) as a controlled, reproducible noise source. While less realistic than the NSTDB noise used in the base paper, this enables precise measurement of SNR improvement against a known noise floor and facilitates fair comparison across all seven benchmarked methods under identical conditions.

6.5 Lightweight Deployable Architecture

Our CNN-BiGRU uses only 2 convolutional layers and 1 biGRU layer (hidden_dim=64), making it significantly lighter than FCN-DAE (12-layer encoder-decoder) and the base paper's biGRU (2 stacked layers). This design prioritises deployability on resource-constrained platforms without sacrificing competitive performance.

6.6 Comparative Results Summary

Method	MSE	SNR (dB)	PRD (%)	Correlation	SNRimp
Classical	—	—	27.07	—	—
Fourier Transform	—	—	20.51	—	—
Wavelet Denoising	—	—	1.21	—	—
DRNN	0.012619	7.85	40.50	0.9441	—
FCN-DAE	0.005703	11.30	27.23	0.9586	—
1D U-Net	0.102537	9.35	37.69	—	+13.96 dB

Method	MSE	SNR (dB)	PRD (%)	Correlation	SNRimp
CNN-BiGRU (Ours)	0.075333	12.35	27.78	—	+16.96 dB

* CNN-BiGRU achieves the best SNR improvement (+16.96 dB) among all DL models and competitive PRD (27.78%) with FCN-DAE (27.23%), while using a significantly simpler architecture.

6.7 Base Paper vs. Our Model — Direct Comparison

Aspect	Dias et al. (biGRU, 2024)	Our CNN-BiGRU
Architecture	biGRU only — no CNN	CNN (k=7,5) + biGRU hybrid
Input	12-lead PTB-XL	Single-lead MIT-BIH
Noise Type	Real: MA, EM, BW (NSTDB)	Synthetic Gaussian (σ=0.5)
SNR Improvement	Up to 30.6 dB	+16.96 dB (controlled SNR)
Baselines Compared	2 (Neurokit only)	6 methods benchmarked
Real-world Validation	Yes — 43 industrial workers	No — synthetic noise only
RMSE	0.023–0.041	0.0753

7. Limitations

7.1 Synthetic Noise Only (No Real Artifacts)

The CNN-BiGRU was trained and tested exclusively on synthetic Gaussian white noise (σ=0.5). Real ECG artifacts — baseline wander, electrode motion, and muscle artifact — have frequency-domain characteristics very different from white noise. The model has not been validated against the NSTDB real physiological noise, which limits confidence in its clinical deployment.

7.2 No Regularization in CNN-BiGRU

Unlike FCN-DAE which employs BatchNormalization after every layer, our CNN-BiGRU has no BatchNorm or Dropout. The gap between training loss (0.00613) and validation loss (0.00617) is small but present, suggesting mild overfitting that may worsen on noisier or out-of-distribution data.

7.3 MSE-Only Loss Function

The model was trained with pure MSE loss, which tends to produce over-smoothed reconstructions. Clinical ECG morphology preservation — particularly the sharpness of R-peaks and the fine structure of P-waves — may be compromised compared to models trained with perceptually-aware losses (e.g., SSD loss as used in FCN-DAE, or gradient-based losses).

7.4 Fixed Noise Amplitude (No SNR Sweep)

All training was conducted at a single noise amplitude (σ=0.5). The model's performance across varying SNR levels (e.g., −10 dB to +10 dB input SNR) has not been characterized, making it impossible to assess robustness in very high or low noise conditions.

7.5 Shallow Recurrent Depth

With only one biGRU layer (hidden_dim=64), the model may be underpowered for complex multi-artifact scenarios. The base paper used 2 stacked biGRU layers, and deeper architectures have shown better performance in related ECG tasks.

7.6 No Real-World Deployment Validation

Unlike Dias et al. who validated on 43 industrial workers in a real-world setting, all evaluations in this project are performed on benchmark database recordings. No clinical or field validation has been conducted, which is a prerequisite for actual medical deployment.

7.7 Wavelet Denoising Dominates on PRD

Despite the architectural innovations of all DL models, Wavelet denoising (PRD: 1.21%) significantly outperforms all deep learning approaches on the PRD metric. This highlights that classical signal processing methods, when well-tuned, remain highly competitive for ECG denoising and that deep learning's primary advantage lies in adaptability and generalizability rather than raw denoising power.

8. Future Recommendations

8.1 Multi-Noise Training

1. Extend the training pipeline to mix Gaussian, baseline wander, muscle artifact, and electrode motion noise using the existing NSTDB processing code (already implemented for DRNN/FCN-DAE). This is the single highest-impact improvement available.

8.2 Architecture Enhancements

2. **BatchNorm + Dropout:** Add BatchNormalization after each CNN layer and a Dropout(0.2) in the FC head to improve regularization.
3. **Deeper biGRU:** Increase biGRU depth to 2 layers with inter-layer dropout — as used in the base paper — to improve temporal modelling capacity.
4. **Combined Loss:** Replace pure MSE loss with a combined MSE + gradient loss (α=0.8) to better preserve R-peak sharpness and fine waveform morphology.

8.3 Attention Mechanism

5. Integrate a lightweight channel or self-attention module (e.g., Squeeze-and-Excitation block or single-head attention) between the CNN and BiGRU stages. This is the key component present in CBGM (Bai et al., 2024) that our model currently lacks, and could meaningfully improve performance without excessive parameter overhead.

8.4 Learning Rate Scheduling

6. Add a ReduceLROnPlateau scheduler (patience=3, factor=0.5) to allow finer convergence. The model is currently saved at epoch 30 (the final epoch), suggesting training had not fully converged.

8.5 SNR Sweep Evaluation

7. Evaluate the trained model at multiple input SNR levels (−10, −5, 0, +5, +10 dB) to produce an SNR improvement curve. This is standard practice in denoising literature and would significantly strengthen the experimental contribution.

8.6 Wavelet-DL Hybrid

8. Explore a pre-processing hybrid: apply Wavelet thresholding as a first-pass denoiser, then feed the residual into the CNN-BiGRU for fine-grained reconstruction. Given Wavelet's PRD of 1.21% vs CNN-BiGRU's 27.78%, this two-stage pipeline could produce the best-of-both results.

8.7 Transfer to Wearable / Edge Deployment

9. Profile the CNN-BiGRU's parameter count, FLOPs, and inference latency, then quantize the model (INT8) for deployment on microcontroller-grade hardware (e.g., ARM Cortex-M, ESP32). The lightweight architecture makes this feasible — a formal deployment benchmark would be a strong contribution.

8.8 Clinical Validation

10. Collaborate with a clinical institution to validate the model on real ambulatory ECG recordings from patients with known arrhythmias. A downstream test — measuring whether denoised ECGs improve arrhythmia classification accuracy — would provide the strongest evidence of clinical utility.

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